Chih-Yuan Chiu

Academic Employment

Georgia Institute of Technology — Aug 2024 - now
Research Engineer II,
School of Electrical and Computer Engineering.
Supervisor: Dr. Justin Romberg

Education

University of California, Berkeley — Sep 2018 - Dec 2023
Ph.D., Department of Electrical Engineering and Computer Sciences
Advisor: Dr. Shankar Sastry

National Taiwan University — Sep 2014 - June 2018
B.S., Department of Electrical Engineering

Contact Information

Emails: cyc@gatech.edu, chihyuanfrankchiu@gmail.com.
Website: `https://chihyuanchiu.github.io/`
Google Scholar: `https://scholar.google.com/citations?hl=en&user=cl9ModoAAAAJ`

Journal Publications

Jihoon Hong, **Chih-Yuan Chiu**, Sara Fridovich-Keil, and Glen Chou. "PolyMerge: Compressing 3D Gaussian Splats with Polytope Coverings for Provably Safe Resource-Constrained Navigation," in *IEEE Robotics and Automation Letters (RA-L),* vol. 11, no. 7, pp. 8512-8519, June 2026.

Zhouyu Zhang*[1], **Chih-Yuan Chiu***, and Glen Chou. "Constraint Learning in Multi-Agent Dynamic Games from Demonstrations of Local Nash Interactions," in *IEEE Robotics and Automation Letters (RA-L),* vol. 11, no. 6, pp. 6696-6703, June 2026.

Chih-Yuan Chiu*, Zhouyu Zhang*, and Glen Chou. "Learning Constraints from Stochastic Partially-Observed Closed-Loop Demonstrations," in *IEEE Control Systems Letters (L-CSS)*, vol. 10, pp. 43-48, January 2026.

Chih-Yuan Chiu and Bryce Ferguson. "Robustness of Incentive Mechanisms Against System Misspecification in Congestion Games," in *IEEE Control Systems Letters (L-CSS)*, vol. 9, pp. 276-281, May 2025.

Chih-Yuan Chiu*, Jingqi Li*, Maulik Bhatt, and Negar Mehr. "To What Extent Do Open-Loop and Feedback Nash Equilibria Diverge in General-Sum Linear Quadratic Dynamic Games?" in *IEEE Control Systems Letters (L-CSS)*, vol. 8, pp. 2583-2588, November 2024.

Lasse Peters, Andrea Bajcsy, **Chih-Yuan Chiu**, David Fridovich-Keil, Forrest Laine, Laura Ferranti, and Javier Alonso-Mora. "Contingency Games for Multi-Agent Interaction," in *IEEE Robotics and Automation Letters (RA-L)*, vol. 9, no. 3, pp. 2208-2215, January 2024. (*Best Paper Award Finalist: TC on Robot Control Best Paper Award*)

Forrest Laine, David Fridovich-Keil, **Chih-Yuan Chiu**, and Claire Tomlin. "The Computation of Approximate Generalized Feedback Nash Equilibria," in *SIAM Journal on Optimization*, vol. 33, no. 1, pp. 294-318, March 2023.

Druv Pai, Michael Psenka, **Chih-Yuan Chiu**, Manxi Wu, Edgar Dobriban, and Yi Ma. "Pursuit of a Discriminative Representation for Multiple Subspaces via Sequential Games," in *Journal of the Franklin Institute*, vol. 360, no. 6, pp. 4135-4171, April 2023.

Amay Saxena*, **Chih-Yuan Chiu***, Ritika Shrivastava, Joseph Menke, and Shankar Sastry. "Simultaneous Localization and Mapping: Through the Lens of Nonlinear Optimization," in *IEEE Robotics and Automation Letters (RA-L)*, vol. 7, no. 3, pp. 7148-7155, July 2022.

[1] The asterisk superscript * indicates equal contribution.

Journal Submissions and Preprints

Chih-Yuan Chiu, Devansh Jalota, and Marco Pavone. "Credit-Based vs. Discount-Based Congestion Pricing: A Comparison Study," arXiv preprint 2602.11077, February 2026. URL: `https://arXiv.org/pdf/2602.11077`. Under review at: *Transportation Research Part B: Methodological*.

Conference Publications

Brendan Gould*, **Chih-Yuan Chiu***, Antoine P. Leeman, Kyriakos G. Vamvoudakis, Samuel Coogan, and Glen Chou. "Over-Approximating Minimizer Sets of Constrained Convex Programs with Parametric Uncertainty via Reachability Analysis," to appear in *IEEE Conference on Decision and Control (CDC),* December 2026.

Chih-Yuan Chiu, Sarah H.Q. Li, and Bryce L. Ferguson. "Distributionally Robust Tolls for Traffic Networks with Affine Latency Functions," to appear in *IEEE Conference on Decision and Control (CDC),* December 2026.

Dan B.W. Choe, Sundhar Vinodh Sangeetha, Steven Emanuel, **Chih-Yuan Chiu**, Samuel Coogan, and Shreyas Kousik. "Ask, Reason, Assist: Robot Collaboration via Natural Language and Temporal Logic," to appear in *IEEE International Conference on Intelligent Robots and Systems (IROS),* September 2026.

Tamer Başar*, Subhonmesh Bose*, Philip Brown*, **Chih-Yuan Chiu***, Bryce L. Ferguson*, Chinmay Maheshwari*, Ketan Savla*, Raj Kiriti Velicheti*, and Manxi Wu*. "Control via Incentives and Information in Socio-technical Systems: A Tutorial," in *American Control Conference (ACC),* pp. 3357-3372, May 2026.

Shuyu Zhan*, **Chih-Yuan Chiu***, Antoine Leeman, and Glen Chou. "Robustly Constrained Dynamic Games for Uncertain Nonlinear Dynamics," in *IEEE International Conference on Robotics and Automation (ICRA),* June 2026.

Sundhar Vinodh Sangeetha, **Chih-Yuan Chiu**, Sarah H.Q. Li, and Shreyas Kousik. "Language Conditioning Improves Accuracy of Aircraft Goal Prediction in Untowered Airspace," in *IEEE International Conference on Robotics and Automation (ICRA),* June 2026. (*Link to Media*).

Zheng Qiu, **Chih-Yuan Chiu**, and Glen Chou. "Active Constraint Learning in High Dimensions from Demonstrations," in *8th Annual Learning for Dynamics & Control Conference (L4DC),* June 2026.

Chih-Yuan Chiu. "Approximately Optimal Toll Design for Efficiency and Equity in Arc-Based Traffic Assignment Models," in *Annual Allerton Conference on Communication, Control, and Computing*, September 2025.

Chih-Yuan Chiu, Devansh Jalota, and Marco Pavone. "Credit vs. Discount-Based Congestion Pricing: A Comparison Study," in *IEEE Conference on Decision and Control (CDC)*, pp. 2331-2336, December 2024.

Chih-Yuan Chiu and Shankar Sastry. "Parameter Estimation in Optimal Tolling for Traffic Networks Under the Markovian Traffic Equilibrium," in *American Control Conference (ACC)*, pp. 1461-1467, July 2024.

Chih-Yuan Chiu, Chinmay Maheshwari, Pan-Yang Su, and Shankar Sastry. "Dynamic Tolling in Arc-based Traffic Assignment Models," in *Annual Allerton Conference on Communication, Control, and Computing*, pp. 1-8, September 2023.

Chih-Yuan Chiu*, Chinmay Maheshwari*, Pan-Yang Su, and Shankar Sastry. "Arc-based Traffic Assignment: Equilibrium Characterization and Learning," in *IEEE Conference on Decision and Control (CDC)*, pp. 7751-7758, December 2023.

Jingqi Li, **Chih-Yuan Chiu**, Lasse Peters, Fernando Palafox, Mustafa Karabag, Javier Alonso-Mora, Somayeh Sojoudi, Claire Tomlin, and David Fridovich-Keil. "Scenario-Game ADMM: A Parallelized Scenario-Based Solver for Stochastic Noncooperative Games," in *IEEE Conference on Decision and Control (CDC)*, pp. 8093-8099, December 2023.

Chih-Yuan Chiu, Kshitij Kulkarni, and Shankar Sastry. "A Novel Statistical Independence Test for Dynamic Causal Discovery with Rare Events," in *IEEE Conference on Decision and Control (CDC)*, pp. 7610-7616, December 2023.

Chih-Yuan Chiu. "SLAM Backends with Objects in Motion: A Unifying Framework and Tutorial," in *American Control Conference (ACC)*, pp. 1635-1642, May 2023.

Jingqi Li, **Chih-Yuan Chiu**, Lasse Peters, Somayeh Sojoudi, Claire Tomlin, and David Fridovich-Keil. "Cost Inference for Feedback Dynamic Games from Noisy Partial State Observations and Incomplete Trajectories," in *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, pp. 1062-1070, June 2023.

Chih-Yuan Chiu and David Fridovich-Keil. "GTP-SLAM: Game-Theoretic Priors for Simultaneous Localization and Mapping in Multi-Agent Scenarios," in *IEEE Conference on Decision and Control (CDC)*, pp. 247-252, December 2022.

Chinmay Maheshwari*, **Chih-Yuan Chiu***, Eric Mazumdar, Shankar Sastry, and Lillian J. Ratliff. "Zeroth-Order Methods for Convex-Concave Minmax Problems: Applications to Decision-Dependent Risk Minimization," in *25th International Conference on Artificial Intelligence and Statistics* (AISTATS), vol. 151, pp. 6702-6734, March 2022.

Chih-Yuan Chiu*, David Fridovich-Keil*, and Claire Tomlin. "Encoding Defensive Driving as a Dynamic Nash Game," in *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 10749-10756, June 2021.

Forrest Laine, David Fridovich-Keil, **Chih-Yuan Chiu**, and Claire Tomlin. "Multi-Hypothesis Interactions in Game-Theoretic Motion Planning," in *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 8016-8023, June 2021.

Forrest Laine, **Chih-Yuan Chiu**, and Claire Tomlin. "Eyes-Closed Safety Kernels: Safety for Autonomous Systems Under Loss of Observability," in *Robotics: Science and Systems (RSS)*, July 2020.

Conference Submissions and Preprints

Chih-Yuan Chiu, and Matthew Hale. "The Cost of Differential Privacy in Linear-Quadratic Dynamic Games." Under review at: *American Controls Conference (ACC) 2027*. URL: `https://arxiv.org/pdf/2610.07238`.

Zhouyu Zhang, **Chih-Yuan Chiu**, and Glen Chou. "Learned Dynamics Enable Optimality Regularizers for Data-Efficient Offline Inverse Constrained RL." Under review at: *International Conference on Learning Representations (ICLR) 2027*.

Theses

(*Ph.D. Thesis*) **Chih-Yuan Chiu**. "Algorithm Design for Safe and Efficient Societal-Scale Navigation," 2023. (`https://www2.eecs.berkeley.edu/Pubs/TechRpts/2023/EECS-2023-267.pdf`).

(*Master's Thesis*) **Chih-Yuan Chiu**. "Simultaneous Localization and Mapping: A Rapprochement of Filtering and Optimization-Based Approaches," 2021. (`https://www2.eecs.berkeley.edu/Pubs/TechRpts/2021/EECS-2021-76.pdf`).

Invited Oral Presentations

(*Upcoming*) **Yue Lab Group Meeting (UMN Aero)** October 2026
"Trustworthy Algorithms for Sustainable Future Mobility"

LEADCAT Group Meeting (UIUC Aero) September 2026
"Trustworthy Algorithms for Sustainable Future Mobility"

IEEE CSS Rising Stars Symposium September 2026
"Trustworthy Algorithms for Sustainable Future Mobility"

MADCAT Group Meeting (Dartmouth Engineering) September 2026
"Trustworthy Algorithms for Sustainable Future Mobility"

3rd MAD-Games Workshop at the 2026 American Control Conference (ACC) May 2026
"Constraint Learning in Multi-Agent Dynamic Games from Demonstrations of Local Nash Interactions"

11th Georgia Tech Postdoctoral Research Symposium Nov 2025
"Credit-based vs. Discount-based Congestion Pricing: A Comparison Study." (Selected as a Top-10 Submission.)

Georgia Tech DCL Symposium Spotlight Talk Apr 2025
"Credit-based vs. Discount-based Congestion Pricing: A Study of Tolling and Toll Relief Mechanisms for Traffic Management."

Georgia Tech ECE Decision and Control Laboratory (DCL) Student Seminar Oct 2024
"Interaction-aware Multi-Agent Control: The Role of Information."

UC Berkeley EECS C106B/206B Guest Lecture Apr 2022
"Simultaneous Localization and Mapping: A Unifying Optimization-Based Framework." (with Ritika Shrivastava)

UC Berkeley Semiautonomous Seminar Feb 2022
"Towards a Rapprochement of Estimation, Prediction, and Planning for Autonomous Navigation."

UC Berkeley EE 221A Guest Lecture Dec 2021
"Simultaneous Localization and Mapping: Filtering and Optimization Approaches." (with Amay Saxena)

UC Berkeley Semiautonomous Seminar Sep 2021
"Gradient-Free Optimistic Gradient Descent-Ascent: Applications to Decision-Dependent Risk Minimization." (with Chinmay Maheshwari)

UC Berkeley Semiautonomous Seminar Jun 2021
"Factor Graphs: A Tool for Optimization-Based Inference in Robotics."

UC Berkeley EECS C106B/206B Guest Lecture Feb 2021
"Simultaneous Localization and Mapping: A Unifying Optimization-Based Framework." (with Amay Saxena)

UC Berkeley Semiautonomous Seminar Jun 2020
"Adversarial-to-Cooperative Games."

Invited Poster Presentations

2025 NSF FRR-NRI Annual Meeting Oct 2025
"Robustly Constrained Dynamic Games with Uncertain Dynamics."

Teaching Experience (Georgia Tech)

Six-Lecture Short Course (Game Theory in Controls) Jul 2025 - Aug 2025
Instructor for a six-lecture short course

Teaching Experience (UC Berkeley)

EE 127 (Optimization Models in Engineering) Aug 2023 - Dec 2023
20-hour Graduate Student Instructor for a 16-week semester-long course

EE 221A (Linear Systems Theory) Aug 2021 - Dec 2021
20-hour Graduate Student Instructor for a 16-week semester-long course

EE 16B (Designing Information Devices and Systems II) Jul 2020 - Aug 2020
25-hour Graduate Student Instructor for an 8-week summer course

Journal and Conference Reviewing

Journals

- IEEE Transactions on Human-Machine Systems.
- IEEE Transactions on Control of Networked Systems.
- IEEE Transactions on Automatic Control.
- IEEE Control Systems Letters.
- IEEE Robotics and Automation Letters (RA-L).
- Journal of Machine Learning Research (JMLR).

Conferences

- IEEE International Conference on Intelligent Transportation Systems (ITSC).
- IEEE Conference on Decision and Control (CDC).
- American Control Conference (ACC).
- IEEE International Conference on Robotics and Automation (ICRA).
- IEEE International Symposium on Multi-Robot and Multi-Agent Systems (MRS).

Research Proposals

(*Under Review*) Google TPU Research & Education Awards 2026: "Scaling Robust Robot Control with TPU Acceleration." Role: **Co-PI**. $100,000.

(*Under Review*) NSF: "Reliable Synthesis and Learning Methods for Safe, Constrained Multi-Agent Interactions in Dynamic Games." Role: **Co-PI**. $600,000.

(*Rejected*) Amazon Research: "Dynamic Closed-Loop Polytope-based Scene Representations for Sustained, Provably Safe, and Resource-Constrained Warehouse Navigation." Role: **PI**. $160,946.

(*Rejected*) Amazon Research: "Reliable Game-Theoretic Motion Planning with Robust Temporal Logic Constraint Satisfaction for Interactive Warehouse Robot Systems." Role: **PI**. $150,000.

(*Rejected*) AI Security Institute: "Non-Bayesian Persuasion for Multi-Agent Autonomy: Leveraging Game-Theoretic Frameworks to Identify and Mitigate Deceptive AI in Engineered Systems." Role: **Co-PI**. $475,000.

Service and Outreach

Ph.D. Thesis Proposal Committee (Mainak Pal, Purdue ECE) September 2026

ACC 2026: Tutorial on Control via Incentives and Information in Socio-technical Systems, Co-Organizer May 2026
Tutorial on Control via Incentives and Information in Socio-technical Systems.

ACC 2026: Workshop on Autonomy in Transportation, Co-Organizer May 2026
Workshop on Emerging Challenges in Multi-agent Planning and Control. (`https://sites.google.com/view/acc2026w05multiagentautonomy/home`)

M.S. Thesis Defense Committee (Maria Cayetana Salinas Rodriguez, ECE) April 2026

M.S. Thesis Defense Committee (Sundhararajan Vinodh Sangeetha) April 2026

Graduate Student Recruitment Committee (GSRC) December 2025 - May 2026
Committee for evaluating Ph.D. applications to Georgia Tech ECE and recommending applicants for first-year Graduate Teaching Assistantships (GTAs).

ACC 2025: Workshop on Mixed-Autonomy Traffic, Co-Organizer July 2025
Workshop on Emerging Challenges and Opportunities in Mixed Autonomy Transportation Systems. (`https://sites.google.com/view/acc2025workshoponmixedautonomy/home?authuser=0`)

M.S. Thesis Proposal Committee (Maria Cayetana Salinas Rodriguez) December 2025

M.S. Thesis Proposal Committee (Sundhararajan Vinodh Sangeetha) November 2025

Decision and Control Laboratory Seminar Co-Organizer Sep 2024 - now
Regular seminar series focused on emerging challenges in the control of robotic and cyber-physical systems at Georgia Tech ECE.

Semiautonomous Seminar Co-Organizer Jan 2020 - Dec 2021
Weekly seminar on control theory, robotics, optimization, computer vision, and machine learning at UC Berkeley EECS.